

A counterexample to a conjecture of A. R. Miller

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Abstract

Let χ be an irreducible character of a finite group G . A. R. Miller conjectured in [6] that the proportion of elements $g \in G$ such that $\chi(g)$ is zero or a root of unity is at least $1/2$. We construct a character of a perfect group of order 69,120 such that this proportion is $511/1152 \approx 0.44$.

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The subject of zeros of characters goes back many years. Given a finite group G , one has its irreducible complex representations, i.e. homomorphisms $\rho: G \rightarrow \mathrm{GL}(V)$, and the associated irreducible characters $\chi(g) = \mathrm{Trace}(\rho(g))$. The number of irreducible representations equals the number of conjugacy classes, and the squares of the degrees satisfy

$$\sum_{\chi \in \mathrm{Irr}(G)} \chi(1)^2 = |G|.$$

The matrices $\rho(g)$ can be quite large, their eigenvalues are roots of unity, and therefore the values $\chi(g)$ are cyclotomic integers: sums of roots of unity, potentially with many terms.

Burnside famously proved that every nonlinear irreducible character χ (i.e. $\chi(1) > 1$) has at least one zero. This follows from an averaging argument together with the orthogonality relation

$$1 = [\chi, \chi] = \frac{1}{|G|} \sum_{g \in G} |\chi(g)|^2.$$

(A recent extension to infinite groups has been given by J. P. Serre in [11].) Later, John G. Thompson refined Burnside’s idea by combining it with a theorem of Siegel, which we recall below, on totally positive algebraic integers (applied to $|\chi(g)|^2$). As a result, for every irreducible character χ of a finite group G , the value $\chi(g)$ is either 0 or a root of unity for more than one third of the elements $g \in G$ (see [3, Problem 3.15]). In other words, these cyclotomic integers $\chi(g)$ —which can a priori be large sums of many roots of unity—are remarkably guaranteed to collapse to 0 or a *single* root of unity on more than a third of the group.

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There has been renewed interest in quantitative and probabilistic vanishing phenomena for character values, that is, questions about the frequency of $\chi(g) = 0$ (or more generally “small” character values) when (χ, g) is chosen at random from $\text{Irr}(G) \times G$ for families such as symmetric groups, classical groups, or finite simple groups ([4], [5]). In [6], Miller proposed an “all finite groups” analogue in which one asks how often an irreducible value collapses to one of the simplest cyclotomic integers: 0 or a root of unity. More precisely, for a finite group G and $\chi \in \text{Irr}(G)$, let

$$\lambda(\chi) := |\{g \in G : \chi(g) = 0 \text{ or } \chi(g) \text{ is a root of unity}\}|.$$

Miller asked [6, Question 1] for the largest constant α such that $\lambda(\chi) \geq \alpha|G|$ holds for every finite G and every $\chi \in \text{Irr}(G)$, and conjectured that $\alpha = 1/2$ [6, Conjecture 1] (see also [7]). He proved this bound for all nilpotent groups, among other cases. This was extended to all groups with a Sylow tower by Moretó–Navarro [9, Theorem C]. Our main point is to give a counterexample to this conjecture showing that α is *strictly smaller* than $1/2$. The conjecture seems to be open for solvable groups, as does [6, Question 1].

Conjecture 1 in [6] also comprised a “dual” conjecture referring to the columns of a character table. This was disproved by Miller [7].

Let us recall now Siegel’s theorem. Let $x \in \mathbb{R} \setminus \{1\}$ be an algebraic integer such that all Galois conjugates $x_1, \dots, x_n$ of x are real and positive. Siegel’s [12, Theorem III] states that

$$\overline{\text{tr}}(x) := \frac{x_1 + \dots + x_n}{n} \geq \frac{3}{2}$$

with equality if and only if $x = \frac{3 \pm \sqrt{5}}{2}$.

Siegel’s theorem has been improved many times, but the exact value of

$$\mu := \liminf_x \overline{\text{tr}}(x)$$

is still unknown. Most recently, Orloski–Sardari–Smith [10, p. 2010] proved that $\mu \geq 1.80203$. On the other hand, the sequence $\{4 \cos(2\pi/p)^2 : p \text{ prime}\}$ shows that $\mu \leq 2$.

Since character values are cyclotomic integers (i.e. integers of a cyclotomic field), it is expected that stronger bounds can be given. In fact, Cassels [1, Lemma 3] proved for every cyclotomic integer x that $\overline{\text{tr}}(|x|^2) \geq 2$ unless x is a sum of at most two roots of unity.

Counterexample. Using GAP [2], we found a perfect group $G \cong C_{12} \cdot (C_2^4 \rtimes A_6)$ of order 69,120 with eight (Galois conjugate) irreducible characters χ of degree 12 such that

$$\frac{\lambda(\chi)}{|G|} = \frac{511}{1152} \approx 0.44.$$

The group can be accessed as $G = \text{PerfectGroup}(69120, 2)$. In particular, Miller’s conjectured bound $\alpha = 1/2$ is false, and the best possible universal constant α satisfies

$$\frac{1}{3} \leq \alpha \leq \frac{511}{1152}.$$

In accordance with Siegel’s and Cassels’s theorems mentioned above, there exists $g \in G$ such that $\chi(g) = \zeta_5 + \overline{\zeta_5} = 2 \cos(2\pi/5)$ and $|\chi(g)|^2 = \frac{3-\sqrt{5}}{2}$ in this example. In fact, there is no $g \in G$ such that $\chi(g)$ is a root of unity. There are a few more such examples among perfect groups, all of them involving A_6 , but we did not find a smaller value of $\lambda(\chi)/|G|$ nor with $\chi(1)$ coprime to 6. It remains an open problem to determine $\min(\lambda(\chi)/|G|)$ over all finite groups.

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